

Simran Mann

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EDUCATION

Simon Fraser University

Bachelor of Applied Science in Computing Science

Sept 2022 – Dec 2026

GPA: 3.99/4.33

- **Relevant Courses:** Distributed Systems, Compilers, Computer Architecture, Operating Systems

TECHNICAL SKILLS

Programming Languages: C/C++ (multithreading, atomics, STL, RAII), Python, x86-64 Assembly, Bash

Systems & Concurrency: POSIX threads, std::thread, mutexes, atomics, MPI, lock-free progress guarantees

Tools & Platforms: Linux/Unix, Git, Docker, LLVM, STM32

EXPERIENCE

SFU Robot Soccer

May 2025 – Present

Firmware Developer

- Develop firmware for a fleet of autonomous robots on ATSAME MCUs using C/C++, focusing on real-time coordination, deterministic behavior, and correctness under tight timing constraints
- Designed basestation firmware to parse commands from a centralized control system over UART and relay them to the robots over LoRa, implementing CRC-based error detection and packet validation
- Implemented real-time motor control and motion primitives (translation and rotation) reasoning about control loops, timing guarantees, and shared resource access to maintain stable robot behavior
- Debugged and validated IMU sensor data over SPI, integrating a Mahony filter and analyzing the impact of noise and sampling behaviour with real data
- Collaborate with the software team on continuous integration and testing to ensure robots follow instructions from an overhead camera-guided brain

NETGEAR

Sep 2024 – April 2025

Orbi Software Test Engineer

- Designed and deployed multi-node network testbeds using managed switches, configuring VLANs, static routing, and network topologies to simulate mesh network environments
- Developed Python automation tool to parse UART serial output, configure devices remotely, and automate network stability testing, reducing test execution time from 8 hours to 2 hours
- Debugged network-level issues including packet loss, routing failures, and inter-device communication problems using Wireshark, tcpdump, ping, traceroute, and iperf for bandwidth analysis
- Validated router and mesh network behavior over TCP/UDP traffic, testing packet forwarding, protocol-level correctness, latency, and failover scenarios

PROJECTS

Parallel and Distributed Graph Algorithms

Sept – Dec 2024

- Implemented parallel PageRank and Triangle Counting algorithms using std::thread for shared-memory systems and MPI for distributed multi-node clusters
- Scaled implementations from single-node multithreaded execution to distributed MPI-based execution, optimizing message-passing to reduce synchronization overhead
- Evaluated vertex-based, edge-based, and dynamic task decomposition approaches to improve load balance and inter-process communication

Concurrent Data Structures

Oct – Nov 2024

- Implemented non-blocking queues, stacks, and sets using lock-free linked lists in C++
- Benchmarked data structures under multi-producer/multi-consumer workloads to analyze contention bottlenecks
- Addressed ABA problems via version-tagged pointers and compare-and-swap (CAS) atomic operations

C Language Compiler

June – Aug 2024

- Developed compiler in C++ using Flex/Bison and LLVM APIs, implementing SSA-form IR code generation and short-circuit evaluation for booleans
- Built a complete pipeline from parsing through semantic analysis to code generation on linux
- Built C runtime and Python scripts to automate testing, validating compilation and execution of code